

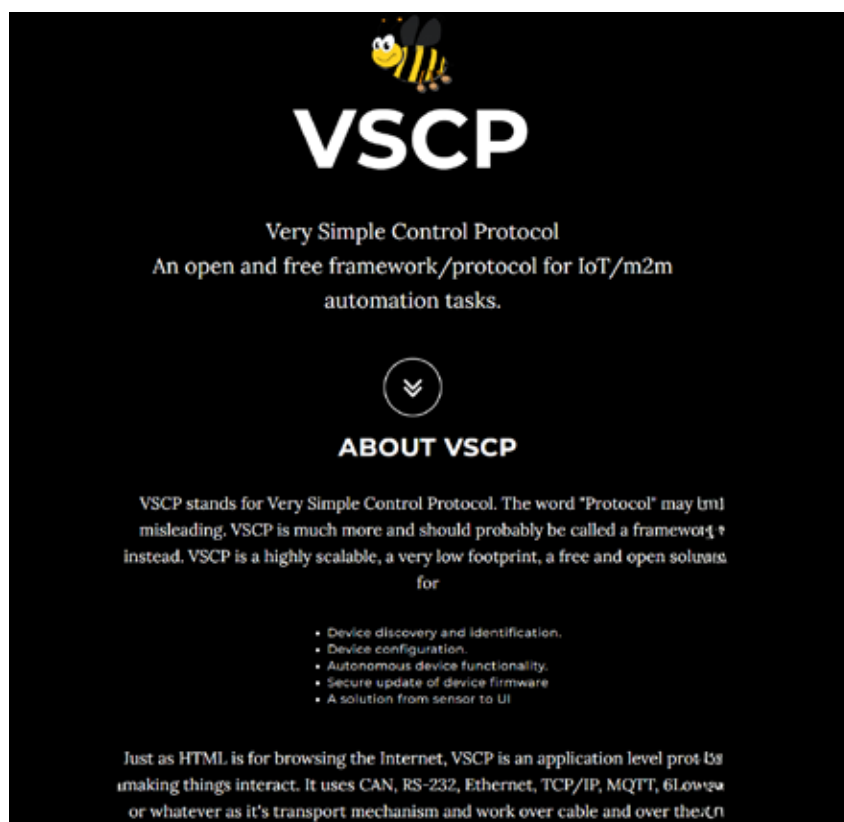


Figure 2: VSCP

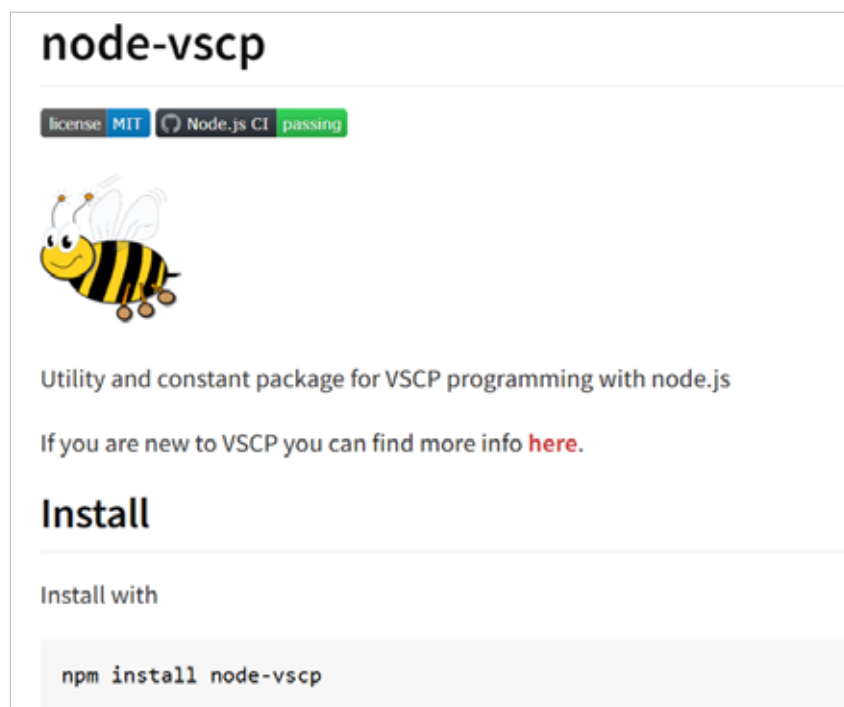


Figure 3: NodeJS package for VSCP

VSCP integration in IoT projects. By leveraging NodeJS programming with Node-VSCP, a very high-performance server-side application can be created that communicates with VSCP networks. NodeJS is ideal for real-time applications because of its async, event-driven model.

To integrate VSCP with NodeRED, the package *node-red-contrib-vscp* is installed so that GUI based scenarios with multiple links can be created. The Node-RED package is available at <https://www.npmjs.com/package/node-red-contrib-vscp>.

Node-VSCP is complemented by an extensive ecosystem as there are assorted packages for HTTP servers, WebSocket communication and to interact with databases. Developers can work on end-to-end IoT solutions that integrate them with dashboards, APIs and other externally accessible services.

Custom development or community nodes help communicate with VSCP from Node-RED. These nodes can help you connect to the VSCP daemon, subscribe to events, and publish commands. In a Node-RED flow they could listen for specific VSCP events (e.g., temperature threshold exceeded) and respond with actions (e.g., send alerts or adjust connected devices, etc). Likewise, VSCP events can be transmitted from Node-RED triggered by actions defined in Node-RED flows. This can provide a powerful environment for IoT automation.

The extensibility of Node-RED also means that VSCP can integrate with other protocols and services, including MQTT, HTTP APIs, cloud services, and many others. All this makes VSCP an excellent interface for building complex IoT ecosystems.

VSCP with Python

Python provides a high-performance ecosystem to interact with the Very Simple Control Protocol (VSCP), enabling developers connect and